

A Packing of 16 Four-Spaces in $\mathbb{R}^{11}$

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Abstract

We give 16 4-dimensional subspaces of $\mathbb{R}^{11}$ whose smallest squared chordal distance is at least 2.714782. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.704184, obtained by deleting 1 of the 17 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 16 subspaces of dimension 4 in $\mathbb{R}^{11}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.714782\dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 11$, $n = 4$. The cell $N = 16$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.704184, obtained by deleting 1 of the 17 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(11 - 4)/11 \cdot 16/15 = 2.715152$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m11n4N16.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 63 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
135739125093622325035816135238330996214145741770689105020251427225446657
899731388026147517978494130294492393379346308353545255408906421312126492
1688488856820102452028052361018594716978202900429
```

denominator:

```
500000000001150065614492309939968800893162243990494795146010851661892109
525145871092016294702119472342807070527788784383129617668093787830037021
331919244116538689455497575564024296721259407488
```

B Coordinates

16 blocks of 11 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m11n4N16.txt`.

```
250537607858 710815864508 -143083206725 -23697422822010026687 -145697304949 175208586985 -29258524206490736688645 -109615208121 309938396183 -409378486329
-18783330939 -312490503911 546624231097 -5150542094457476117090 591473008613 -32111680481 -630089864788185457064 -108954232127 -165279479018 383850264061
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